THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL MCQ TEST SET 3

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. Two angles of a linear pair differ by 40°. The smaller angle is: 1

(a) 70° **(b)** 80° **(c)** 60° **(d)** 90°

2. For every whole number n, $n + (n+1) + (n+2)$ is always a multiple of: 1

(a) 2 **(b)** 6 **(c)** 3 **(d)** 4

3. Sides 13 cm, 6 cm and 8 cm are given. Which statement is true? 1

(a) only a right triangle is possible **(b)** only an equilateral triangle is possible

(c) no triangle can be formed **(d)** a triangle can be formed

4. $\frac{5}{7}$ of $2\frac{4}{5}$ is: 1

(a) $3\frac{18}{35}$ **(b)** 2 **(c)** 4 **(d)** 1

5. If 1 lakh = 1,00,000, how many lakhs are there in 3,40,00,000? 1

(a) 350 **(b)** 340 **(c)** 341 **(d)** 339

6. A cyclist covers 28 km in the first hour and 3 km less every following hour for 4 hours in all. The total distance (km) is: 1

(a) 99 **(b)** 94 **(c)** 93 **(d)** 95

7. What must be added to 0.875 to get 2? 1

(a) 1.025 **(b)** 1.225 **(c)** 1.125 **(d)** 2.125

8. If $x = 3$ and $y = 6$, the value of $5x - 4y + xy$ is: 1

(a) 8 **(b)** 11 **(c)** 10 **(d)** 9

9. One angle of a linear pair is 3 times the other. The larger angle is: 1

(a) 145° **(b)** 135° **(c)** 155° **(d)** 125°

10. If $1 + 2 + 3 + \cdots + n = 406$, then n is: 1

(a) 29 **(b)** 28 **(c)** 30 **(d)** 27

11. In an isosceles triangle with perimeter 50 cm the equal sides are 18 cm each. The base (cm) is: 1

(a) 15 **(b)** 13 **(c)** 14 **(d)** 16

12. If $x \times \frac{2}{5} = \frac{4}{15}$, then $x \div \frac{2}{3}$ is: 1

(a) $\frac{1}{2}$ **(b)** $1\frac{1}{2}$ **(c)** 1 **(d)** 2

13. The number of digits in the product 99,999 × 9,999 is: 1

(a) 11 **(b)** 10 **(c)** 8 **(d)** 9

14. The value of $(100 - (4 \times (12 - 7) + 10)) \div (2 \times 5)$ is: 1

(a) 5 **(b)** 70 **(c)** 7 **(d)** 9

15. Which fraction is equal to 0.375? 1

(a) $\frac{3}{7}$ **(b)** $\frac{3}{5}$ **(c)** $\frac{3}{8}$ **(d)** $\frac{3}{80}$

16. A rule is: output $= 3\times$ input -7. The output for the input 12 is: 1

(a) 28 **(b)** 30 **(c)** 31 **(d)** 29

17. An angle is equal to its supplement. The angle is: 1

(a) 180° **(b)** 45° **(c)** 90° **(d)** 60°

18. The number of reverse-and-add steps for 79 to become a palindrome is: 1

(a) 7 **(b)** 5 **(c)** 8 **(d)** 6

19. The two acute angles of a right triangle are in the ratio $2:3$. The smaller acute angle is: 1

(a) 26° **(b)** 46° **(c)** 36° **(d)** 56°

20. $\frac{2}{3} \div (\frac{4}{5} \div \frac{2}{5})$ equals: 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{6}$ **(c)** $2\frac{1}{12}$ **(d)** $\frac{2}{3}$

21. How many lakhs are there in 35 crore? 1

(a) 3510 **(b)** 3500 **(c)** 4500 **(d)** 3490

22. The value of $96 \div (4 \times (7-3)) + 5 \times (11-8)$ is: 1

(a) 23 **(b)** 212 **(c)** 21 **(d)** 1862

23. Which of these is the greatest? 1

(a) $\frac{3}{4}$ **(b)** 0.85 **(c)** 0.8 **(d)** $\frac{7}{8}$

24. If $x = 5$, the value of $3x^2 + 4$ is: 1

(a) 81 **(b)** 80 **(c)** 79 **(d)** 78

25. Two supplementary angles are in the ratio $4:5$. The larger angle is: 1

(a) 110° **(b)** 100° **(c)** 120° **(d)** 90°

26. Which expression is a multiple of 11 for every digit pair (a, b)? 1

(a) $10a + b$ **(b)** $(10a + b) + (10b + a)$ **(c)** $a - b$ **(d)** $(10a + b) - (10b + a)$

27. The number of isosceles triangles with whole-number sides and perimeter 15 cm (equilateral included) is: 1

(a) 6 **(b)** 4 **(c)** 5 **(d)** 3

28. $\frac{4}{9}$ of a number is 20. Three times the number is: 1

(a) $67\frac{1}{2}$ **(b)** $135\frac{1}{2}$ **(c)** 270 **(d)** 135

29. The number of seconds in 100 days is: 1

(a) 86,36,400 **(b)** 86,40,000 **(c)** 86,41,000 **(d)** 8,64,000

30. The value of $48 \div (2 \times (5+1)) \times (14-6)$ is: 1

(a) 128 **(b)** 32 **(c)** 1688 **(d)** 34

31. $0.5 \times 0.5 \times 0.8 =$ 1

(a) 2 **(b)** 0.2 **(c)** 20 **(d)** 0.02

32. If $a = 5$ and $b = 2$, the value of $2a^2 - 9b$ is: 1

(a) 31 **(b)** 33 **(c)** 34 **(d)** 32

33. The supplement of the complement of 50° is: 1

(a) 140° **(b)** 160° **(c)** 130° **(d)** 150°

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34. The smallest 5-digit number divisible by 9 is: 1

(a) 10009 **(b)** 10007 **(c)** 10008 **(d)** 10017

35. The number of triangles with whole-number sides (in cm) and perimeter 14 cm is: 1

(a) 3 **(b)** 6 **(c)** 5 **(d)** 4

36. A pipe fills $\frac{5}{6}$ of a tank in 20 minutes. The time to fill the whole tank (minutes) is: 1

(a) $24\frac{1}{2}$ **(b)** 12 **(c)** 48 **(d)** 24

37. The number of zeros in 4,500 × 2,000 is: 1

(a) 7 **(b)** 8 **(c)** 5 **(d)** 6

38. The value of 'twice the sum of 15 and 4, decreased by 11' is: 1

(a) 38 **(b)** 27 **(c)** 16 **(d)** 49

39. How many times does 0.04 go into 1.2? 1

(a) 29 **(b)** 31 **(c)** 30 **(d)** 32

40. Simplify $6(x+1)-2(x-4)$: 1

(a) $4x+2$ **(b)** $8x+14$ **(c)** $4x+14$ **(d)** $4x-2$

41. **Assertion (A):** Vertically opposite angles are equal. 1

Reason (R): Vertically opposite angles always add up to 180°.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

42. **Assertion (A):** The sum of a two-digit number and its reverse is divisible by 11. 1

Reason (R): (10a + b) + (10b + a) = 10(a + b).

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

43. **Assertion (A):** The exterior angle of a triangle equals the sum of the two interior opposite angles. 1

Reason (R): The exterior angle of a triangle always equals the adjacent interior angle.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

44. **Assertion (A):** Dividing by $\frac{2}{3}$ is the same as multiplying by $\frac{3}{2}$. 1

Reason (R): Dividing by a fraction is the same as multiplying by the fraction itself.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

45. **Assertion (A):** 10 lakh has 5 zeros. 1
Reason (R): 1 lakh has 5 zeros.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** $12 \div 3 \times 2 = 2$. 1
Reason (R): Multiplication and division are done from left to right.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** 0.3 x 0.2 = 0.6. 1
Reason (R): The product of two decimals has as many decimal places as the two factors have in all.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** 4x + 3y = 7xy. 1
Reason (R): Only like terms can be added to give a single term.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** When a transversal cuts two parallel lines, the corresponding angles are supplementary. 1
Reason (R): When a transversal cuts two parallel lines, the corresponding angles are equal.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** The product of two odd numbers is even. 1
Reason (R): The product of an even number and any whole number is even.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**